

Pollution mapping for LSAP and dioceses

Master methods sources and map documentation

6 October 2026

This master report brings together the twelve pollution maps and thirteen layouts in LSAP_Pollution.aprx. It records the sources, executed analysis workflows, output fields, cartography and interpretation limits for annual air pollution, monitored river chemistry, river plastic outfalls and mapped landfill inventories. The analyses provide environmental context around dioceses and LSAP communities; they do not establish personal exposure or causal responsibility.

All analyses use copies of 2,874 dioceses and 2,739 LSAP member points. Copies of 122 episcopal conference features and 194 country features preserve ecclesiastical context. Stable DIO_ID links outputs to the copied boundaries. Original foundational data were not edited (LSAP Pollution Atlas, 2026, foundation_manifest.json).

World layouts use Robinson (EPSG 54030). Analysis coordinates are WGS84 (EPSG 4326), with EPSG 6933 used for equal area footprint operations and geodesic methods for global distance and polygon area. Land_km2 names the full polygon area, not a separately masked land area. Member positional precision has not been independently verified.

Map inventory

Map	Subject	Units
P01	Fine particulate air pollution across dioceses	$\mu\text{g}/\text{m}^3$
P02	Annual nitrogen dioxide across dioceses	$10^{15} \text{ molecules}/\text{cm}^2$
P03	Nitrate in monitored diocesan rivers	mg/L as nitrogen
P04	Phosphorus in monitored diocesan rivers	mg/L
P05	Oxygen in monitored diocesan rivers	mg/L dissolved oxygen
P06	River plastic outflows and LSAP communities	Metric tonnes/year
P07	LSAP communities near river plastic outfalls	Kilometres
P08	Mapped waste sites across dioceses	Distinct sites
P09	Mapped landfill footprints across dioceses	Hectares
P10	LSAP communities near mapped landfills	Kilometres
P11	Annual PM25 pollution surface	$\mu\text{g}/\text{m}^3$
P12	Annual NO2 atmospheric composite	$10^{15} \text{ molecules}/\text{cm}^2$

Shared analysis and preservation methods

1. Copy dioceses, LSAP members, conferences and countries into FoundationCopies.gdb. Preserve ecclesiastical attributes; leave foundational source datasets untouched.
2. Project analysis copies into WGS84. Assign stable DIO_ID; calculate geodesic polygon area. Use Robinson for world display, independently of analysis coordinates.
3. Inspect source CRS, dimensions, units, cell size, NoData and latitude extent before each raster analysis. PM_{2.5} is a 0.01° geographic grid spanning 60°S–70°N; NO₂ and valid-month count are 0.2° geographic grids. None is equal area.
4. For concentration rasters, calculate native cell geodesic area and fractional diocesan boundary overlap. Calculate $\Sigma(\text{concentration} \times \text{overlap area}) / \Sigma(\text{valid overlap area})$; calculate area-weighted spatial median. Boundary fractions are measured in the native geographic grid and multiplied by geodesic cell area, an approximation within boundary cells.
5. Calculate coverage against the full diocesan polygon, including areas outside the raster extent. Retain statistics but shade only dioceses with at least 50% valid coverage. This threshold is a display rule, not a source quality certification.
6. Create layouts, popups, layer files and map files. Use the requested pollution palette; grey indicates missing or insufficient evidence; teal identifies foundation member dots.

Simple unweighted zonal means would overweight smaller geographic cells at high latitudes. Equal-area cells remove that particular bias, but fractional boundary overlap and valid coverage still matter. Land_km2 is the full polygon geodesic area, despite its legacy field name; it is not separately masked to terrestrial land.

Freshwater retention and source attribution

The selected GFQA v3 archive is the openly published CC BY 4.0 dataset (Heinle et al., 2026). Raw freshwater observations, station coordinates and downloaded scratch files were removed from this pollution project after aggregation. Source metadata, citations, QA counts and workflows remain. This removal follows the requested output design; it is not an independent legal determination. Waste source layers, regional raw extracts and derivatives remain. © OpenStreetMap contributors; ODbL.

Validation and interpretation limits

project_QA.json records twelve maps, thirteen layouts, no broken layers and no raw freshwater connections. Freshwater reporting thresholds and nonnegative member distances passed the scripted checks. Completed ArcGIS map exports were visually inspected during map construction. The Word reports have structural validation; final pagination could not be checked because the document renderer is unavailable on this machine.

Waste Atlas supplementation is not part of these results. The retained OSM inventory excludes point-only dumps and has uneven coverage and source dates. Source grids and filtering parameters are audited before raster aggregation; no unweighted geographic grid zonal mean is used. PM25 fill values were validated before mapping and statistics. No combined pollution severity index is calculated across the different units or periods.

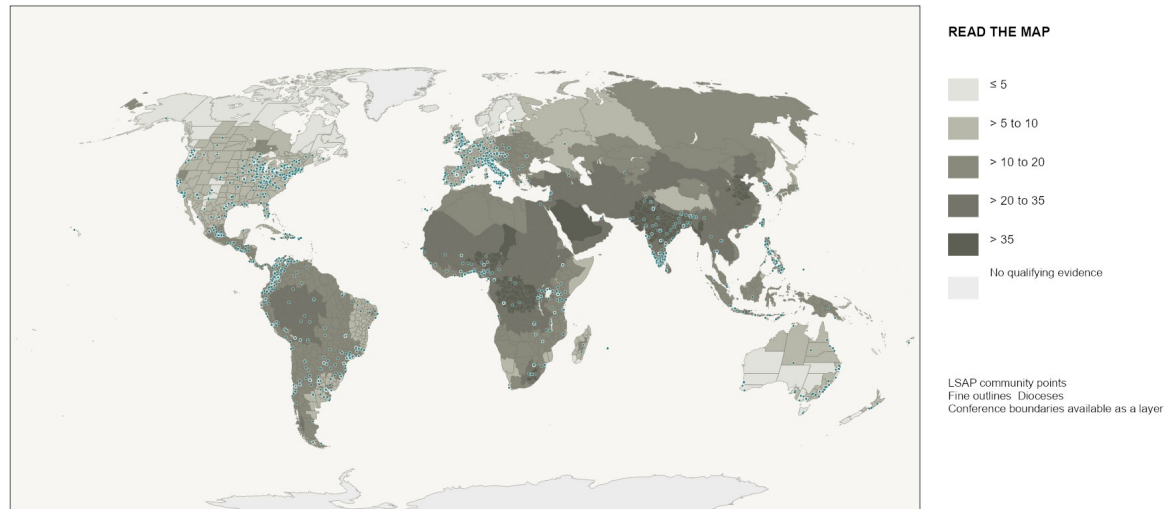
P01 Fine particulate air pollution across dioceses

This report explains the source data, executed workflow and interpretation of fine particulate air pollution across dioceses. It documents the existing ArcGIS Pro output, including its coverage limits and retained ecclesiastical context. (ACAG, n.d.; Shen et al., 2024).

Map P01 | Documentation date 6 October 2026 | Units $\mu\text{g}/\text{m}^3$ | Air year 2021

Fine particulate air pollution across dioceses

Area weighted annual PM2.5 · 2021 · $\mu\text{g}/\text{m}^3$



Shading requires 50% valid polygon coverage. Full statistics remain in popups. This is ambient context, not personal exposure.

SatPM V6GL03 · CC BY 4.0 · Native 0.01° grid

LSAP POLLUTION ATLAS / COPIED ECCLESIASTICAL AND MEMBER FOUNDATIONS / P01

Figure P01 World overview exported from the completed ArcGIS Pro layout.

Shading requires 50% valid polygon coverage. Full statistics remain in popups. This is ambient context, not personal exposure.

All analyses use copies of 2,874 dioceses and 2,739 LSAP member points. Copies of 122 episcopal conference features and 194 country features preserve ecclesiastical context. Stable DIO_ID links outputs to the copied boundaries. Original foundational data were not edited (LSAP Pollution Atlas, 2026, foundation_manifest.json).

Analysis workflow

The following numbered sequence describes the executed ArcPy and Python analyses as a typed ModelBuilder style workflow. Custom raster, station aggregation and geometry operations are script steps, not a claim that an ArcGIS ModelBuilder model file was run. Source data: (ACAG, n.d.; Shen et al., 2024).

1. Download SatPM V6GL03 CNN global annual 2021 NetCDF; preserve source and attribution.
2. Read native latitude/longitude coordinates and PM25 variable; export a float GeoTIFF on its native grid, units $\mu\text{g}/\text{m}^3$. Convert negative scaled source fill values to NoData in PM25_validated.tif before aggregation and display.
3. Apply the shared weighted raster aggregation to each diocesan copy; retain AirMean, AirMedian, ValidAreaPct, ValidCells and AirStory. Correct latitude-extent coverage against the full polygon.
4. Set AirDisplay to AirMean when valid area coverage is at least 50%; otherwise leave null. Join all ecclesiastical context through the copied polygon attributes.
5. Classify diocesan AirDisplay at 5, 10, 20 and 35 $\mu\text{g}/\text{m}^3$. Overlay copied LSAP dots, diocesan outlines and optional conference boundaries.

Area weighting and coverage

Both concentration rasters use geographic cells, not equal area cells. For diocesan aggregation: weighted mean = $\sum(c_i \times a_i) / \sum(a_i)$, summing only valid cells. a_i is geodesic row cell area multiplied by native grid overlap fraction. This treats concentration as constant within a cell; geographic boundary fractions approximate true geodesic overlap within boundary cells. Spatial median uses the same area weights.

Coverage uses the full polygon denominator, including the PM25 extent correction at 60°S–70°N. AirDisplay requires at least 50 percent valid coverage while all computed statistics remain in attributes. The NO2 implementation uses global native grid polygon overlap for its denominator; PM25 partial extent correction uses the ratio of clipped to full geodesic polygon area. These are documented numerical approximations, not population weighting.

Output fields and files

Field or band	Meaning
AirMean	Area weighted mean over valid overlap
AirMedian	Area weighted spatial median
ValidAreaPct	Valid overlap as percent of full diocesan polygon
ValidCells	Number of contributing native raster cells
AirDisplay	Mean where coverage is at least 50 percent; otherwise null
AirStory	Interpretation narrative and coverage
AirYear and AirUnits	2021 and concentration units

Analysis dataset: PollutionAnalysis.gdb/Dioceses_PM25. Styles/P01.lyrx and Styles/P01.mapx preserve layer and map definitions. Map_Exports/P01_World_overview.png preserves the layout preview. The editable project is LSAP_Pollution.aprx.

Retained raster audit reports 2,871 dioceses meeting the 50 percent shading threshold. These counts differ from the number with any computed statistics (LSAP Pollution Atlas, 2026, raster_audit.json).

Interpretation and data handling

2021 is the selected common air comparison year. Class boundaries support cartography; they do not classify annual health compliance. Source: SatPM global annual file.

Credit the Atmospheric Composition Analysis Group for the SatPM V6GL03 data, and retain the source and version citation when sharing the map. The source acquisition record identifies CC BY 4.0. Source files and the validated raster copy remain in the project.

Cartographic classes are descriptive display intervals, not regulatory or ecological quality grades. The pollution ramp is #E1E2DB, #B7B8A9, #898A7C, #74746A and #5D5E54. Lower oxygen and shorter proximity distances use the dark end. Foundation member dots use teal #247C80; proximity maps color member points by distance.

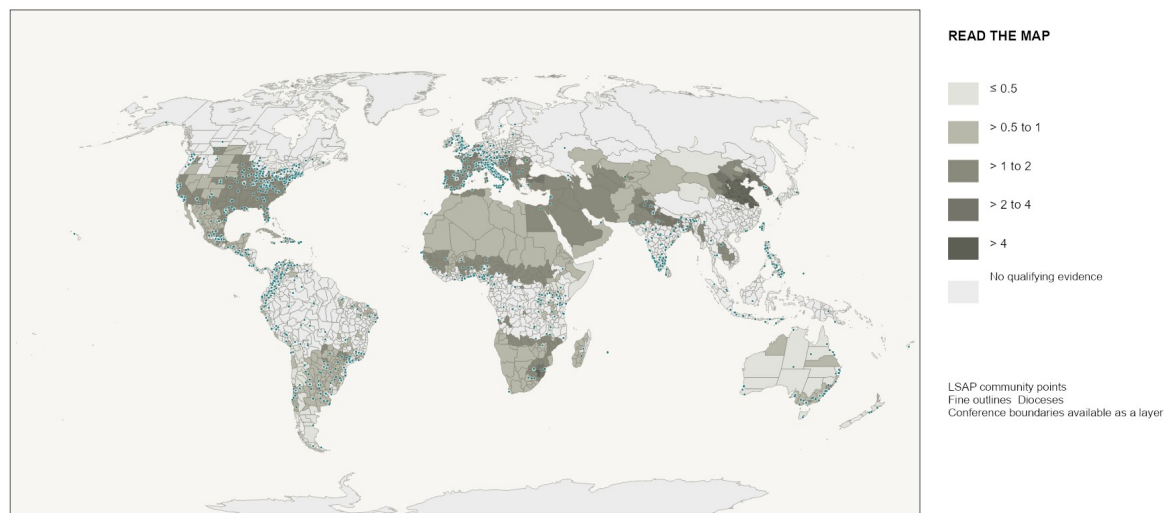
P02 Annual nitrogen dioxide across dioceses

This report explains the source data, executed workflow and interpretation of annual nitrogen dioxide across dioceses. It documents the existing ArcGIS Pro output, including its coverage limits and retained ecclesiastical context. (Glissenaar et al., 2024, 2025; KNMI, n.d.).

Map P02 | Documentation date 6 October 2026 | Units 10^{15} molecules/cm² | Air year 2021

Annual nitrogen dioxide across dioceses

Area weighted tropospheric NO₂ - 2021 - 10^{15} molecules/cm²



KNMI ESA CCI+ TROPOMI L3 - v1.26 - 0.2° grid

LSAP POLLUTION ATLAS / COPIED ECCLESIASTICAL AND MEMBER FOUNDATIONS / P02

Figure P02 World overview exported from the completed ArcGIS Pro layout.

Shading requires 50% valid polygon coverage. Monthly means weighted by calendar days. At least 9 qualifying months and all quarters. Column density differs from ground-level exposure.

All analyses use copies of 2,874 dioceses and 2,739 LSAP member points. Copies of 122 episcopal conference features and 194 country features preserve ecclesiastical context. Stable DIO_ID links outputs to the copied boundaries. Original foundational data were not edited (LSAP Pollution Atlas, 2026, foundation_manifest.json).

Analysis workflow

The following numbered sequence describes the executed ArcPy and Python analyses as a typed ModelBuilder style workflow. Custom raster, station aggregation and geometry operations are script steps, not a claim that an ArcGIS ModelBuilder model file was run. Source data: (Glissenaar et al., 2024, 2025; KNMI, n.d.).

1. Download all twelve 2021 KNMI ESA CCI+ TROPOMI L3 v1.26 monthly files (fv1.0 filename version), native 900 × 1800 grid.
2. Select monthly cells with qa_L3=1, land/transition mask, finite column value and fractional effective monthly coverage × calendar days ≥ 10. This effective coverage is not a literal count of valid days.
3. Calculate calendar-day-weighted means of qualifying monthly columns. Require at least nine qualifying months and representation from all four quarters. Preserve negative retrieval values; divide by 10^{15} to express molecules/cm² in readable units.

4. Export annual concentration and qualifying-month-count rasters; audit both CRS and grids. Aggregate annual concentration using geodesic cell areas and fractional overlaps. Retain weighted spatial median and full-polygon valid coverage.
5. Classify diocesan AirDisplay at 0.5, 1, 2 and 4×10^{15} molecules/cm². Overlay copied LSAP dots, diocesan outlines and optional conference boundaries.

Area weighting and coverage

Both concentration rasters use geographic cells, not equal area cells. For diocesan aggregation: weighted mean = $\sum(c_i \times a_i) / \sum(a_i)$, summing only valid cells. a_i is geodesic row cell area multiplied by native grid overlap fraction. This treats concentration as constant within a cell; geographic boundary fractions approximate true geodesic overlap within boundary cells. Spatial median uses the same area weights.

Coverage uses the full polygon denominator, including the PM25 extent correction at 60°S–70°N. AirDisplay requires at least 50 percent valid coverage while all computed statistics remain in attributes. The NO2 implementation uses global native grid polygon overlap for its denominator; PM25 partial extent correction uses the ratio of clipped to full geodesic polygon area. These are documented numerical approximations, not population weighting.

Output fields and files

Field or band	Meaning
AirMean	Area weighted mean over valid overlap
AirMedian	Area weighted spatial median
ValidAreaPct	Valid overlap as percent of full diocesan polygon
ValidCells	Number of contributing native raster cells
AirDisplay	Mean where coverage is at least 50 percent; otherwise null
AirStory	Interpretation narrative and coverage
AirYear and AirUnits	2021 and concentration units

Analysis dataset: PollutionAnalysis.gdb/Dioceses_NO2. Styles/P02.lyrx and Styles/P02.mapx preserve layer and map definitions. Map_Exports/P02_World_overview.png preserves the layout preview. The editable project is LSAP_Pollution.aprx.

Retained raster audit reports 1,404 dioceses meeting the 50 percent shading threshold. These counts differ from the number with any computed statistics (LSAP Pollution Atlas, 2026, raster_audit.json).

Interpretation and data handling

A cell with nine qualifying months is an observed-month estimate, not complete annual sampling. Tropospheric column density is different from surface concentration or personal exposure. KNMI source.

Cartographic classes are descriptive display intervals, not regulatory or ecological quality grades. The pollution ramp is #E1E2DB, #B7B8A9, #898A7C, #74746A and #5D5E54. Lower oxygen and shorter proximity distances use the dark end. Foundation member dots use teal #247C80; proximity maps color member points by distance.

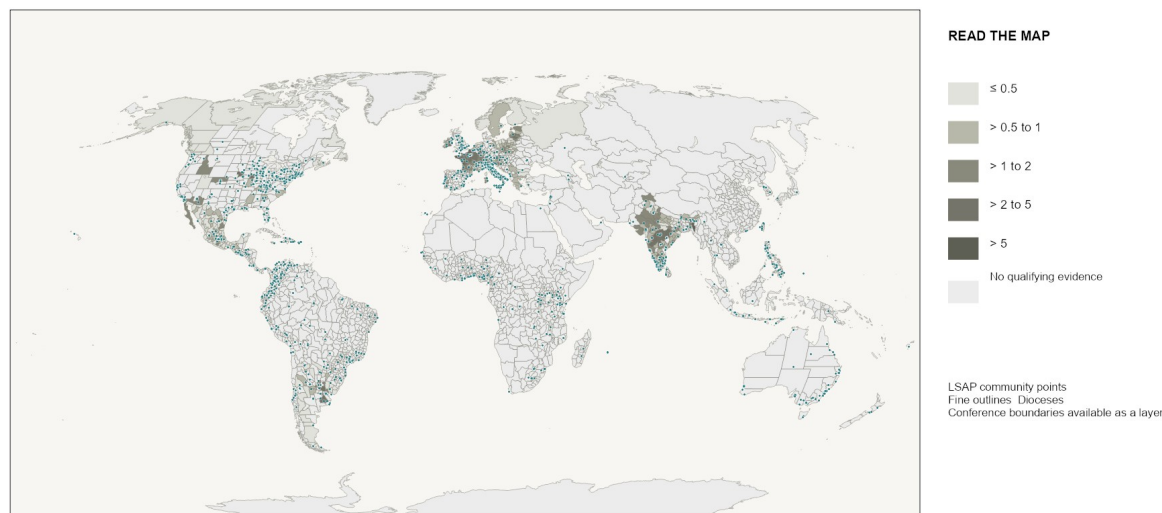
P03 Nitrate in monitored diocesan rivers

This report explains the source data, executed workflow and interpretation of nitrate in monitored diocesan rivers. It documents the existing ArcGIS Pro output, including its coverage limits and retained ecclesiastical context. (Heinle et al., 2026).

Map P03 | Documentation date 6 October 2026 | Units mg/L as nitrogen | River observation window 2015–2024

Nitrate in monitored diocesan rivers

Median of river station year medians · 2015–2024 · mg/L as N



At least 3 stations and 5 station-years. Censored values excluded. Uneven monitoring; grey means insufficient or missing evidence.

UNEP GEMS/Water GFQA v3 · CC BY 4.0 · doi:10.5281/zenodo.18459694

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Figure P03 World overview exported from the completed ArcGIS Pro layout.

At least 3 stations and 5 station-years. Censored values excluded. Uneven monitoring; grey means insufficient or missing evidence.

All analyses use copies of 2,874 dioceses and 2,739 LSAP member points. Copies of 122 episcopal conference features and 194 country features preserve ecclesiastical context. Stable DIO_ID links outputs to the copied boundaries. Original foundational data were not edited (LSAP Pollution Atlas, 2026, foundation_manifest.json).

Analysis workflow

The following numbered sequence describes the executed ArcPy and Python analyses as a typed ModelBuilder style workflow. Custom raster, station aggregation and geometry operations are script steps, not a claim that an ArcGIS ModelBuilder model file was run. Source data: (Heinle et al., 2026).

1. Download GFQA v3 to a dedicated temporary FreshwaterScratch folder. Read station and measurement records from the archive in memory; create no persistent station layer.
2. Select river stations, observations from 2015–2024, the exact parameter and mg/L units. Exclude flagged censored values, bad/poor/suspect quality, negatives and nonfinite results.
3. Assign stations to copied dioceses using point-in-polygon; deterministic smallest DIO_ID resolves overlapping/boundary matches.
4. Calculate each station-year median, then the diocesan median and interquartile range of station-year medians. Retain counts of distinct stations, station-years and monitored years.

5. Publish only when at least three stations and five station-years qualify. Suppress derived numeric values for smaller groups; retain all diocesan polygons and narrative coverage status.
6. Symbolize Median using boundaries 0.5, 1, 2 and 5 mg/L as N. Reverse the palette for dissolved oxygen so lower oxygen is darker. Create diocesan story popups.
7. Verify only aggregated freshwater feature classes are referenced by the project; remove the downloaded archive and all scratch freshwater source records. Retain aggregate QA, source attribution and reproducible scripts.

Output fields and files

Field or band	Meaning
Median	Median of qualifying station year medians
IQR	Interquartile range of station year medians
Stations and StationYears	Distinct stations and station year count
Years	Distinct monitored years in selected window
Coverage	Reporting status rather than area coverage
WaterStory	Monitoring and interpretation narrative
Parameter Units Period	Parameter identity, mg/L and 2015–2024

Analysis dataset: PollutionAnalysis.gdb/Freshwater_NO3N. Styles/P03.lyrx and Styles/P03.mapx preserve layer and map definitions. Map_Exports/P03_World_overview.png preserves the layout preview. The editable project is LSAP_Pollution.aprx.

630 dioceses meet reporting thresholds; 357,971 measurements were accepted. All 2,874 copied dioceses are retained; missing and suppressed summaries remain null (LSAP Pollution Atlas, 2026, freshwater_aggregate_QA.json).

Interpretation and data handling

These statistics describe monitored rivers, not every waterbody. Censored-value exclusion can bias concentrations upward. Station-year medians give each station-year equal weight; they are not area-weighted water surfaces. GFQA v3, CC BY 4.0. Raw-data removal follows the user's output requirement; deletion alone does not establish compliance with every possible use condition.

FreshwaterScratch/GFQA_v3.zip and the extracted method metadata were deleted after validating the aggregate outputs and project connections. No raw station observations or station geometry layer remains in this pollution project. Aggregate QA, nonobservational source metadata and scripts are retained. CC BY 4.0 attribution remains required; source removal was an output requirement, not a claim that deletion alone determines licence compliance (Heinle et al., 2026; LSAP Pollution Atlas, 2026, cleanup_audit.json).

Cartographic classes are descriptive display intervals, not regulatory or ecological quality grades. The pollution ramp is #E1E2DB, #B7B8A9, #898A7C, #74746A and #5D5E54. Lower oxygen and shorter proximity distances use the dark end. Foundation member dots use teal #247C80; proximity maps color member points by distance.

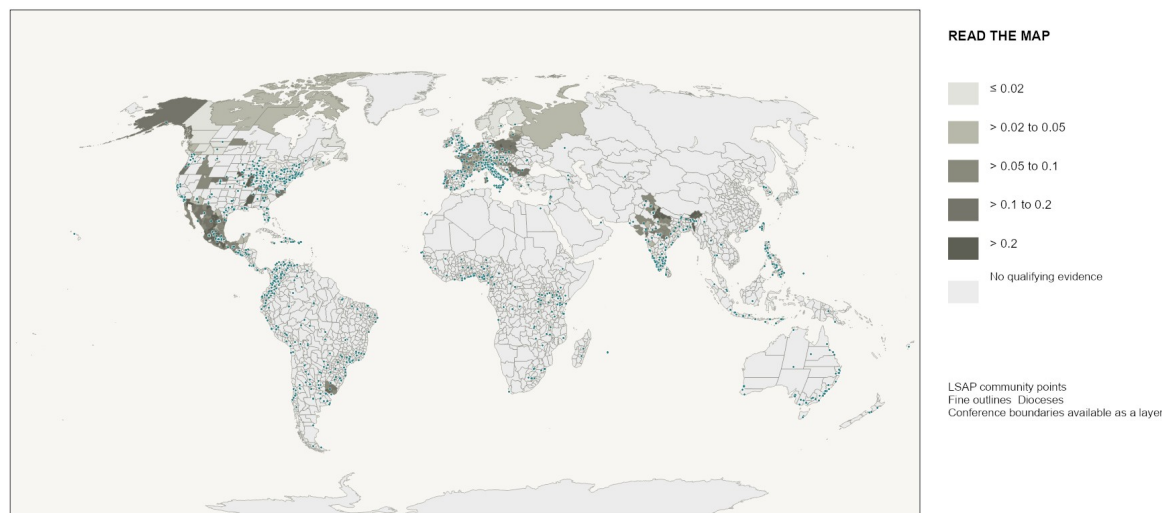
P04 Phosphorus in monitored diocesan rivers

This report explains the source data, executed workflow and interpretation of phosphorus in monitored diocesan rivers. It documents the existing ArcGIS Pro output, including its coverage limits and retained ecclesiastical context. (Heinle et al., 2026).

Map P04 | Documentation date 6 October 2026 | Units mg/L | River observation window 2015–2024

Phosphorus in monitored diocesan rivers

Median of river station year medians · 2015–2024 · mg/L



River observations only. These summaries describe monitored sites rather than all waters. Raw records are removed from the final workspace.

UNEP GEMS/Water GFQA v3 · CC BY 4.0

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Figure P04 World overview exported from the completed ArcGIS Pro layout.

River observations only. These summaries describe monitored sites rather than all waters. Raw records are removed from the final workspace.

All analyses use copies of 2,874 dioceses and 2,739 LSAP member points. Copies of 122 episcopal conference features and 194 country features preserve ecclesiastical context. Stable DIO_ID links outputs to the copied boundaries. Original foundational data were not edited (LSAP Pollution Atlas, 2026, foundation_manifest.json).

Analysis workflow

The following numbered sequence describes the executed ArcPy and Python analyses as a typed ModelBuilder style workflow. Custom raster, station aggregation and geometry operations are script steps, not a claim that an ArcGIS ModelBuilder model file was run. Source data: (Heinle et al., 2026).

1. Download GFQA v3 to a dedicated temporary FreshwaterScratch folder. Read station and measurement records from the archive in memory; create no persistent station layer.
2. Select river stations, observations from 2015–2024, the exact parameter and mg/L units. Exclude flagged censored values, bad/poor/suspect quality, negatives and nonfinite results.
3. Assign stations to copied dioceses using point-in-polygon; deterministic smallest DIO_ID resolves overlapping/boundary matches.
4. Calculate each station-year median, then the diocesan median and interquartile range of station-year medians. Retain counts of distinct stations, station-years and monitored years.

5. Publish only when at least three stations and five station-years qualify. Suppress derived numeric values for smaller groups; retain all diocesan polygons and narrative coverage status.
6. Symbolize Median using boundaries 0.02, 0.05, 0.1 and 0.2 mg/L. Reverse the palette for dissolved oxygen so lower oxygen is darker. Create diocesan story popups.
7. Verify only aggregated freshwater feature classes are referenced by the project; remove the downloaded archive and all scratch freshwater source records. Retain aggregate QA, source attribution and reproducible scripts.

Output fields and files

Field or band	Meaning
Median	Median of qualifying station year medians
IQR	Interquartile range of station year medians
Stations and StationYears	Distinct stations and station year count
Years	Distinct monitored years in selected window
Coverage	Reporting status rather than area coverage
WaterStory	Monitoring and interpretation narrative
Parameter Units Period	Parameter identity, mg/L and 2015–2024

Analysis dataset: PollutionAnalysis.gdb/Freshwater_TP. Styles/P04.lyrx and Styles/P04.mapx preserve layer and map definitions. Map_Exports/P04_World_overview.png preserves the layout preview. The editable project is LSAP_Pollution.aprx.

512 dioceses meet reporting thresholds; 326,811 measurements were accepted. All 2,874 copied dioceses are retained; missing and suppressed summaries remain null (LSAP Pollution Atlas, 2026, freshwater_aggregate_QA.json).

Interpretation and data handling

These statistics describe monitored rivers, not every waterbody. Censored-value exclusion can bias concentrations upward. Station-year medians give each station-year equal weight; they are not area-weighted water surfaces. GFQA v3, CC BY 4.0. Raw-data removal follows the user's output requirement; deletion alone does not establish compliance with every possible use condition.

FreshwaterScratch/GFQA_v3.zip and the extracted method metadata were deleted after validating the aggregate outputs and project connections. No raw station observations or station geometry layer remains in this pollution project. Aggregate QA, nonobservational source metadata and scripts are retained. CC BY 4.0 attribution remains required; source removal was an output requirement, not a claim that deletion alone determines licence compliance (Heinle et al., 2026; LSAP Pollution Atlas, 2026, cleanup_audit.json).

Cartographic classes are descriptive display intervals, not regulatory or ecological quality grades. The pollution ramp is #E1E2DB, #B7B8A9, #898A7C, #74746A and #5D5E54. Lower oxygen and shorter proximity distances use the dark end. Foundation member dots use teal #247C80; proximity maps color member points by distance.

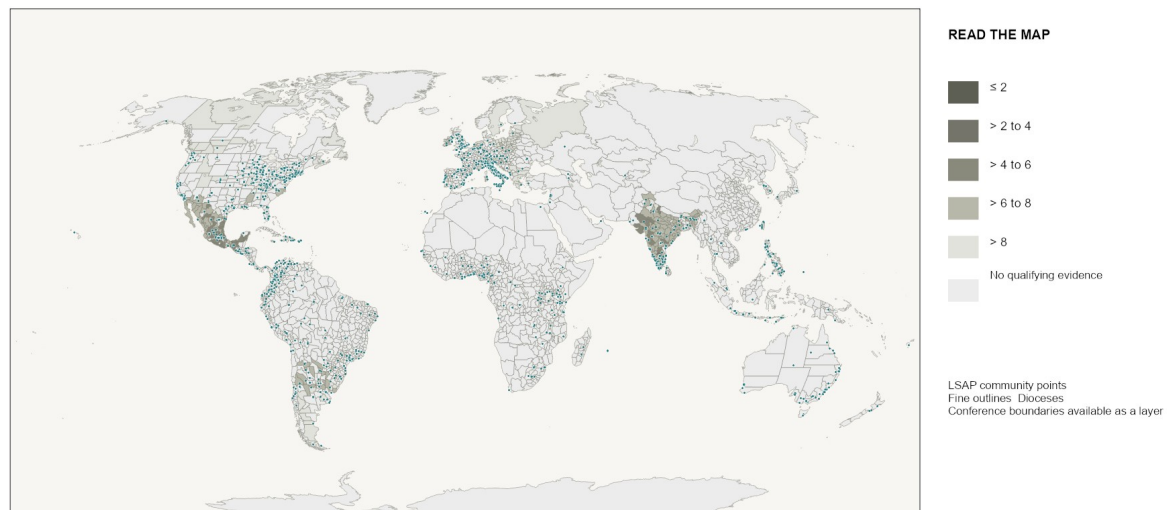
P05 Oxygen in monitored diocesan rivers

This report explains the source data, executed workflow and interpretation of oxygen in monitored diocesan rivers. It documents the existing ArcGIS Pro output, including its coverage limits and retained ecclesiastical context. (Heinle et al., 2026).

Map P05 | Documentation date 6 October 2026 | Units mg/L dissolved oxygen | River observation window 2015–2024

Oxygen in monitored diocesan rivers

Median of river station year medians · 2015–2024 · mg/L



Darker shades indicate lower dissolved oxygen. This map does not assign a universal ecological quality grade.

UNEP GEMS/Water GFOA v3 · CC BY 4.0

LSAP POLLUTION ATLAS / COPIED ECCLESIASTICAL AND MEMBER FOUNDATIONS / P05

Figure P05 World overview exported from the completed ArcGIS Pro layout.

Darker shades indicate lower dissolved oxygen. This map does not assign a universal ecological quality grade.

All analyses use copies of 2,874 dioceses and 2,739 LSAP member points. Copies of 122 episcopal conference features and 194 country features preserve ecclesiastical context. Stable DIO_ID links outputs to the copied boundaries. Original foundational data were not edited (LSAP Pollution Atlas, 2026, foundation_manifest.json).

Analysis workflow

The following numbered sequence describes the executed ArcPy and Python analyses as a typed ModelBuilder style workflow. Custom raster, station aggregation and geometry operations are script steps, not a claim that an ArcGIS ModelBuilder model file was run. Source data: (Heinle et al., 2026).

1. Download GFQA v3 to a dedicated temporary FreshwaterScratch folder. Read station and measurement records from the archive in memory; create no persistent station layer.
2. Select river stations, observations from 2015–2024, the exact parameter and mg/L units. Exclude flagged censored values, bad/poor/suspect quality, negatives and nonfinite results.
3. Assign stations to copied dioceses using point-in-polygon; deterministic smallest DIO_ID resolves overlapping/boundary matches.
4. Calculate each station-year median, then the diocesan median and interquartile range of station-year medians. Retain counts of distinct stations, station-years and monitored years.

5. Publish only when at least three stations and five station-years qualify. Suppress derived numeric values for smaller groups; retain all diocesan polygons and narrative coverage status.
6. Symbolize Median using boundaries 2, 4, 6 and 8 mg/L. Reverse the palette for dissolved oxygen so lower oxygen is darker. Create diocesan story popups.
7. Verify only aggregated freshwater feature classes are referenced by the project; remove the downloaded archive and all scratch freshwater source records. Retain aggregate QA, source attribution and reproducible scripts.

Output fields and files

Field or band	Meaning
Median	Median of qualifying station year medians
IQR	Interquartile range of station year medians
Stations and StationYears	Distinct stations and station year count
Years	Distinct monitored years in selected window
Coverage	Reporting status rather than area coverage
WaterStory	Monitoring and interpretation narrative
Parameter Units Period	Parameter identity, mg/L and 2015–2024

Analysis dataset: PollutionAnalysis.gdb/Freshwater_O2_Dis. Styles/P05.lyrx and Styles/P05.mapx preserve layer and map definitions. Map_Exports/P05_World_overview.png preserves the layout preview. The editable project is LSAP_Pollution.aprx.

544 dioceses meet reporting thresholds; 294,224 measurements were accepted. All 2,874 copied dioceses are retained; missing and suppressed summaries remain null (LSAP Pollution Atlas, 2026, freshwater_aggregate_QA.json).

Interpretation and data handling

These statistics describe monitored rivers, not every waterbody. Censored-value exclusion can bias concentrations upward. Station-year medians give each station-year equal weight; they are not area-weighted water surfaces. GFQA v3, CC BY 4.0. Raw-data removal follows the user’s output requirement; deletion alone does not establish compliance with every possible use condition.

FreshwaterScratch/GFQA_v3.zip and the extracted method metadata were deleted after validating the aggregate outputs and project connections. No raw station observations or station geometry layer remains in this pollution project. Aggregate QA, nonobservational source metadata and scripts are retained. CC BY 4.0 attribution remains required; source removal was an output requirement, not a claim that deletion alone determines licence compliance (Heinle et al., 2026; LSAP Pollution Atlas, 2026, cleanup_audit.json).

Cartographic classes are descriptive display intervals, not regulatory or ecological quality grades. The pollution ramp is #E1E2DB, #B7B8A9, #898A7C, #74746A and #5D5E54. Lower oxygen and shorter proximity distances use the dark end. Foundation member dots use teal #247C80; proximity maps color member points by distance.

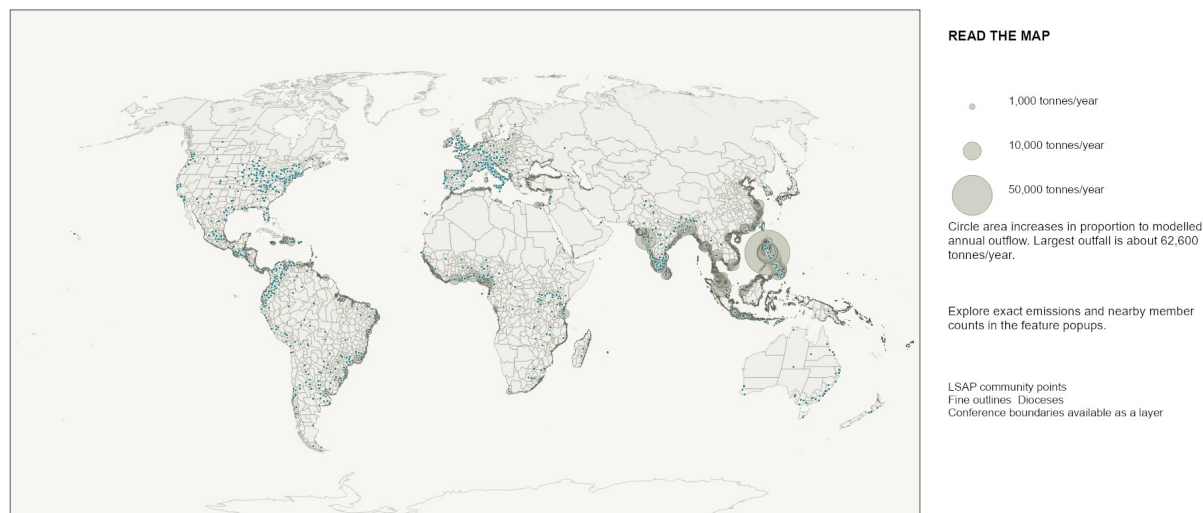
P06 River plastic outflows and LSAP communities

This report explains the source data, executed workflow and interpretation of river plastic outflows and LSAP communities. It documents the existing ArcGIS Pro output, including its coverage limits and retained ecclesiastical context. (Meijer et al., 2021, data set and article).

Map P06 | Documentation date 6 October 2026 | Units Metric tonnes/year | Model published 2021

River plastic outflows and LSAP communities

Modelled annual macroplastic outflow · midpoint scenario · tonnes/year



Circle area is proportional to emission. Teal dots are LSAP members. Distances use the supplied member coordinates, whose precision is unverified.

Meijer et al. 2021 · doi:10.6084/m9.figshare.14515590 · CC BY 4.0

LSAP POLLUTION ATLAS / COPIED ECCLESIASTICAL AND MEMBER FOUNDATIONS / P06

Figure P06 World overview exported from the completed ArcGIS Pro layout.

Circle area is proportional to emission. Teal dots are LSAP members. Distances use the supplied member coordinates, whose precision is unverified.

All analyses use copies of 2,874 dioceses and 2,739 LSAP member points. Copies of 122 episcopal conference features and 194 country features preserve ecclesiastical context. Stable DIO_ID links outputs to the copied boundaries. Original foundational data were not edited (LSAP Pollution Atlas, 2026, foundation_manifest.json).

Analysis workflow

The following numbered sequence describes the executed ArcPy and Python analyses as a typed ModelBuilder style workflow. Custom raster, station aggregation and geometry operations are script steps, not a claim that an ArcGIS ModelBuilder model file was run. Source data: (Meijer et al., 2021, data set and article).

1. Download and preserve the Meijer et al. GIS shapefile and metadata; copy 31,819 outfalls into the analysis geodatabase.
2. Copy dots_exten into Emission_t_yr: midpoint modelled annual floating macroplastic emission in metric tonnes/year.
3. Generate geodesic near tables between copied LSAP member points and outfalls. Summarize distinct member counts within 20 km and 100 km for each outfall; join counts to the source copy.
4. Use proportional circles with area proportional to Emission_t_yr; use the pollution palette and translucent fills with dark edges. Overlay teal LSAP dots and ecclesiastical outlines.
5. Create world and Southeast Asia layouts with popups showing exact emissions and member counts.

Output fields and files

Field or band	Meaning
Emission_t_yr	Midpoint annual macroplastic emission; copied from dots_exten
Members20 and Members100	Distinct LSAP members within 20 and 100 km

Analysis dataset: PollutionAnalysis.gdb/River_Plastic_Outfalls. Styles/P06.lyrx and Styles/P06.mapx preserve layer and map definitions. Map_Exports/P06_World_overview.png preserves the layout preview. The editable project is LSAP_Pollution.aprx.

An additional Southeast Asia detail layout and P06_Southeast_Asia_detail.png show the relation between proportional emission circles and member dots. Circle diameter is $40 \times \sqrt{\text{emission} / \text{maximum emission}}$ points; circle area is proportional to emission.

Interpretation and data handling

The model estimates emissions to the ocean at outfalls. Geographic proximity is not watershed connectivity, causal contribution or exposure. Meijer GIS dataset, CC BY 4.0.

The Meijer midpoint GIS data are CC BY 4.0. Retain the authors, data DOI, scenario and units with any reused source or derived map. The original shapefile and copied outfall layer remain in the project.

Cartographic classes are descriptive display intervals, not regulatory or ecological quality grades. The pollution ramp is #E1E2DB, #B7B8A9, #898A7C, #74746A and #5D5E54. Lower oxygen and shorter proximity distances use the dark end. Foundation member dots use teal #247C80; proximity maps color member points by distance.

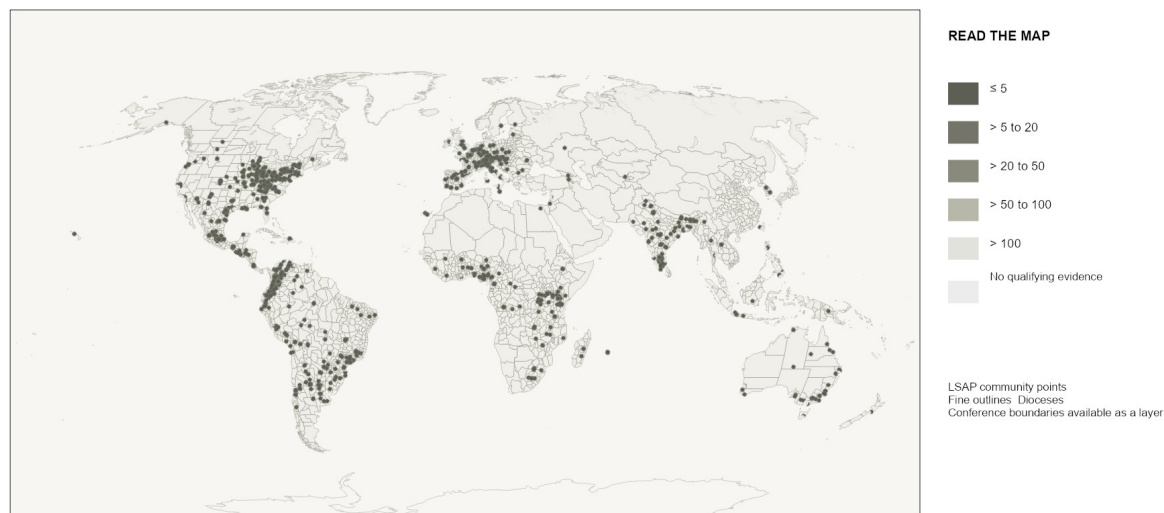
P07 LSAP communities near river plastic outfalls

This report explains the source data, executed workflow and interpretation of lsap communities near river plastic outfalls. It documents the existing ArcGIS Pro output, including its coverage limits and retained ecclesiastical context. (Meijer et al., 2021, data set and article).

Map P07 | Documentation date 6 October 2026 | Units Kilometres | Model published 2021

LSAP communities near river plastic outfalls

Nearest river outfall - geodesic distance in km



Meijer et al. 2021 - copied LSAP member foundations

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Figure P07 World overview exported from the completed ArcGIS Pro layout.

Proximity is straight-line geographic distance. A nearby outfall may belong to another watershed.

All analyses use copies of 2,874 dioceses and 2,739 LSAP member points. Copies of 122 episcopal conference features and 194 country features preserve ecclesiastical context. Stable DIO_ID links outputs to the copied boundaries. Original foundational data were not edited (LSAP Pollution Atlas, 2026, foundation_manifest.json).

Analysis workflow

The following numbered sequence describes the executed ArcPy and Python analyses as a typed ModelBuilder style workflow. Custom raster, station aggregation and geometry operations are script steps, not a claim that an ArcGIS ModelBuilder model file was run. Source data: (Meijer et al., 2021, data set and article).

1. Copy LSAP points into Member_Plastic_Proximity; preserve foundational member fields.
2. Generate geodesic nearest-outfall distances; join outfall identifier and modelled annual emission.
3. Summarize outfalls within 20 km and 100 km for every member.
4. Symbolize nearest distance at 5, 20, 50 and 100 km; nearby members receive darker pollution colors. Retain all dioceses and optional conference boundaries.
5. Create member popups with nearest distance, emission and counts.

Output fields and files

Field or band	Meaning
NearOut_ID and Near_km	Nearest outfall identifier and geodesic distance
Near_t_yr	Modelled emission of nearest outfall
Outfalls20 and Outfalls100	Outfall counts within 20 and 100 km
DistanceNote	Interpretation and positional precision note

Analysis dataset: PollutionAnalysis.gdb/Member_Plastic_Proximity. Styles/P07.lyrx and Styles/P07.mapx preserve layer and map definitions. Map_Exports/P07_World_overview.png preserves the layout preview. The editable project is LSAP_Pollution.aprx.

Interpretation and data handling

Distances use supplied member coordinates; positional precision has not been independently verified. Nearby outfalls may drain another watershed.

The Meijer midpoint GIS data are CC BY 4.0. Retain the authors, data DOI, scenario and units with any reused source or derived map. The original shapefile and copied outfall layer remain in the project.

Cartographic classes are descriptive display intervals, not regulatory or ecological quality grades. The pollution ramp is #E1E2DB, #B7B8A9, #898A7C, #74746A and #5D5E54. Lower oxygen and shorter proximity distances use the dark end. Foundation member dots use teal #247C80; proximity maps color member points by distance.

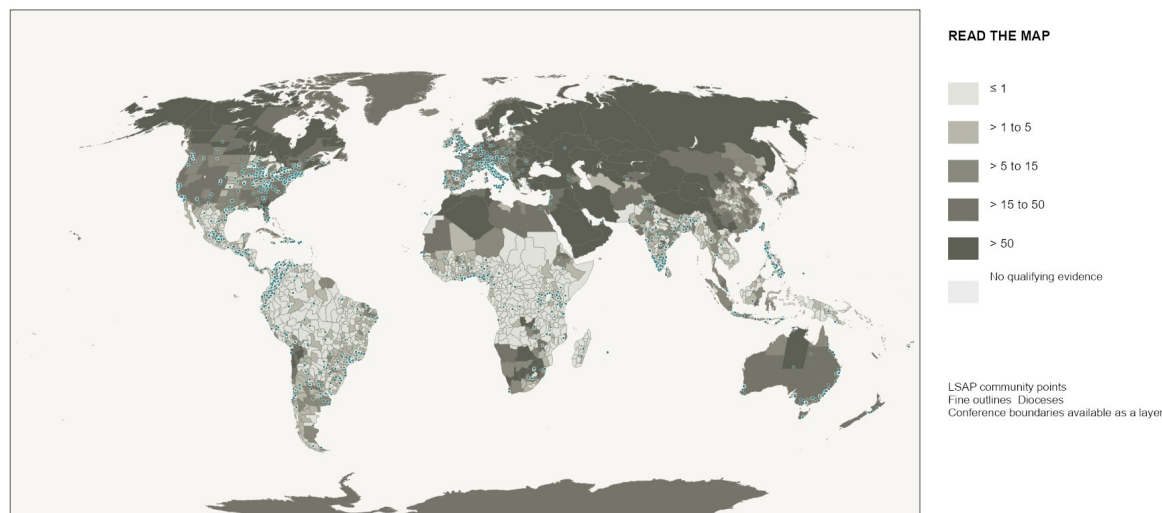
P08 Mapped waste sites across dioceses

This report explains the source data, executed workflow and interpretation of mapped waste sites across dioceses. It documents the existing ArcGIS Pro output, including its coverage limits and retained ecclesiastical context. (OpenStreetMap contributors, 2026; OpenStreetMap Wiki contributors, n.d.).

Map P08 | Documentation date 6 October 2026 | Units Distinct sites | Inventory retrieved 6 October 2026

Mapped waste sites across dioceses

Distinct mapped landfill sites assigned to each diocese



© OpenStreetMap contributors · ODbL · regional mirrors retrieved 6 October 2026

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Figure P08 World overview exported from the completed ArcGIS Pro layout.

Each site is assigned once by greatest footprint overlap. Inventory coverage varies. Zero means no mapped sites, not verified absence.

All analyses use copies of 2,874 dioceses and 2,739 LSAP member points. Copies of 122 episcopal conference features and 194 country features preserve ecclesiastical context. Stable DIO_ID links outputs to the copied boundaries. Original foundational data were not edited (LSAP Pollution Atlas, 2026, foundation_manifest.json).

Analysis workflow

The following numbered sequence describes the executed ArcPy and Python analyses as a typed ModelBuilder style workflow. Custom raster, station aggregation and geometry operations are script steps, not a claim that an ArcGIS ModelBuilder model file was run. Source data: (OpenStreetMap contributors, 2026; OpenStreetMap Wiki contributors, n.d.).

1. Query landuse=landfill polygons from seven public regional OpenStreetMap land-use mirrors; retain original JSON, service provenance and the copied OSM_Landfill_Source layer.
2. Deduplicate stable OSM identifiers, exclude missing geometry and repair a working copy. Retain 51,748 unique polygon sites; control status and pollution severity remain unknown.
3. Intersect footprints with copied dioceses using EPSG:6933 equal-area geometry. Assign each distinct site once to the diocese with greatest footprint overlap; resolve ties deterministically.
4. Summarize assigned distinct SiteCount; separately retain IntersectSites for all touching dioceses. Join counts to every diocesan polygon and calculate sites per 1,000 km².

5. Symbolize counts at 1, 5, 15 and 50; retain working footprints for local detail and original source in the geodatabase.

Output fields and files

Field or band	Meaning
SiteCount	Sites uniquely assigned by greatest footprint overlap
IntersectSites	Distinct footprints touching the diocese
Footprint_ha	Unioned clipped footprint area in hectares
AreaPct	Footprint area as percent of full geodesic polygon area
Sites1000km2	Assigned sites per 1000 square kilometres
WasteStory	Partial inventory narrative

Analysis dataset: PollutionAnalysis.gdb/Dioceses_Landfills. Styles/P08.lyrx and Styles/P08.mapx preserve layer and map definitions. Map_Exports/P08_World_overview.png preserves the layout preview. The editable project is LSAP_Pollution.aprx.

Interpretation and data handling

Partial polygon inventory; point-only dumps are excluded. Zero means no mapped site. Waste Atlas has not been incorporated: downloadable records and redistribution permission were not verified. Regional source freshness varies, including an older European mirror.

The working inventory contains 51,748 unique sites after removing 2,529 duplicate records. Of these, 50,996 receive a primary diocese assignment and 752 remain unassigned. Raw regional JSON, OSM_Landfill_Source and Landfills_Working remain available. Site control status and pollution severity are unknown. The European mirror is older than most other regions; retrieval date does not imply a uniform data vintage (LSAP Pollution Atlas, 2026, waste_QA.json).

Cartographic classes are descriptive display intervals, not regulatory or ecological quality grades. The pollution ramp is #E1E2DB, #B7B8A9, #898A7C, #74746A and #5D5E54. Lower oxygen and shorter proximity distances use the dark end. Foundation member dots use teal #247C80; proximity maps color member points by distance.

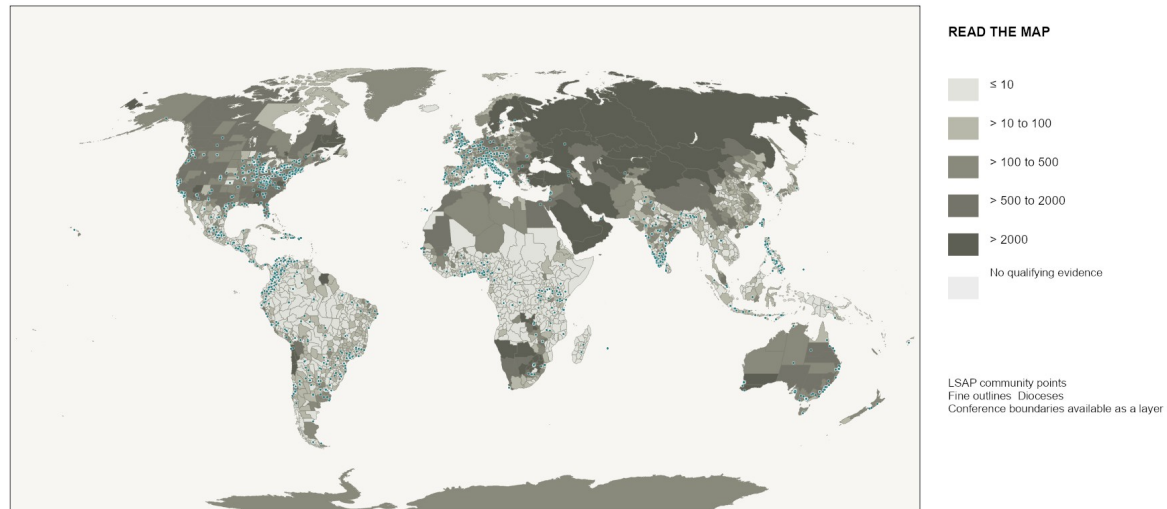
P09 Mapped landfill footprints across dioceses

This report explains the source data, executed workflow and interpretation of mapped landfill footprints across dioceses. It documents the existing ArcGIS Pro output, including its coverage limits and retained ecclesiastical context. (OpenStreetMap contributors, 2026; OpenStreetMap Wiki contributors, n.d.).

Map P09 | Documentation date 6 October 2026 | Units Hectares | Inventory retrieved 6 October 2026

Mapped landfill footprints across dioceses

Unioned mapped landfill footprint · hectares within diocese



Footprints split at diocesan boundaries; overlapping footprint area counted once. Generic landfill tags do not establish pollution severity.

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Figure P09 World overview exported from the completed ArcGIS Pro layout.

Footprints split at diocesan boundaries; overlapping footprint area counted once. Generic landfill tags do not establish pollution severity.

All analyses use copies of 2,874 dioceses and 2,739 LSAP member points. Copies of 122 episcopal conference features and 194 country features preserve ecclesiastical context. Stable DIO_ID links outputs to the copied boundaries. Original foundational data were not edited (LSAP Pollution Atlas, 2026, foundation_manifest.json).

Analysis workflow

The following numbered sequence describes the executed ArcPy and Python analyses as a typed ModelBuilder style workflow. Custom raster, station aggregation and geometry operations are script steps, not a claim that an ArcGIS ModelBuilder model file was run. Source data: (OpenStreetMap contributors, 2026; OpenStreetMap Wiki contributors, n.d.).

1. Use repaired working footprint copies and copied diocesan boundaries.
2. Intersect footprints and boundaries in equal-area EPSG:6933; union clipped footprints by diocese so overlapping mapped polygons count once.
3. Calculate unioned Footprint_ha; divide by full geodesic diocesan polygon area to calculate AreaPct. Retain site counts alongside area metrics.
4. Join to all copied dioceses; classify footprint hectares at 10, 100, 500 and 2,000. Expose AreaPct and count metrics in popups.

5. Keep original source, working footprints and derivative summaries; credit © OpenStreetMap contributors and ODbL.

Output fields and files

Field or band	Meaning
SiteCount	Sites uniquely assigned by greatest footprint overlap
IntersectSites	Distinct footprints touching the diocese
Footprint_ha	Unioned clipped footprint area in hectares
AreaPct	Footprint area as percent of full geodesic polygon area
Sites1000km2	Assigned sites per 1000 square kilometres
WasteStory	Partial inventory narrative

Analysis dataset: PollutionAnalysis.gdb/Dioceses_Landfills. Styles/P09.lyrx and Styles/P09.mapx preserve layer and map definitions. Map_Exports/P09_World_overview.png preserves the layout preview. The editable project is LSAP_Pollution.aprx.

Interpretation and data handling

Mapped footprint area is not contaminated area. A generic landfill tag does not distinguish controlled landfill from open dump. OSM tag definition.

The working inventory contains 51,748 unique sites after removing 2,529 duplicate records. Of these, 50,996 receive a primary diocese assignment and 752 remain unassigned. Raw regional JSON, OSM_Landfill_Source and Landfills_Working remain available. Site control status and pollution severity are unknown. The European mirror is older than most other regions; retrieval date does not imply a uniform data vintage (LSAP Pollution Atlas, 2026, waste_QA.json).

Cartographic classes are descriptive display intervals, not regulatory or ecological quality grades. The pollution ramp is #E1E2DB, #B7B8A9, #898A7C, #74746A and #5D5E54. Lower oxygen and shorter proximity distances use the dark end. Foundation member dots use teal #247C80; proximity maps color member points by distance.

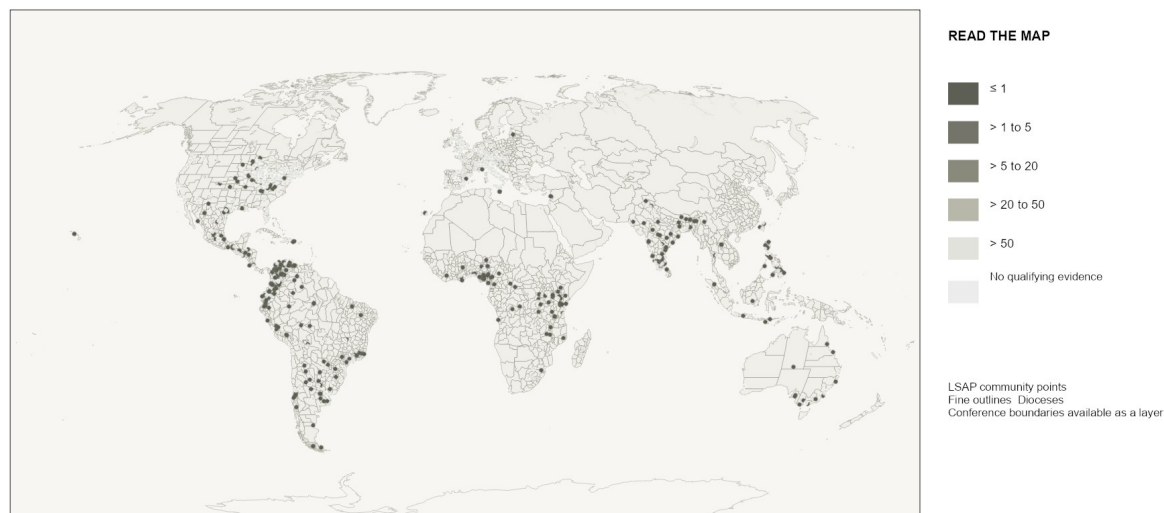
P10 LSAP communities near mapped landfills

This report explains the source data, executed workflow and interpretation of lsap communities near mapped landfills. It documents the existing ArcGIS Pro output, including its coverage limits and retained ecclesiastical context. (OpenStreetMap contributors, 2026; OpenStreetMap Wiki contributors, n.d.).

Map P10 | Documentation date 6 October 2026 | Units Kilometres | Inventory retrieved 6 October 2026

LSAP communities near mapped landfills

Nearest mapped landfill footprint - geodesic distance in km



Distance to polygon edge; zero if inside. Site presence and proximity do not establish contamination. Member precision is unverified.

© OpenStreetMap contributors - ODbL - copied LSAP foundations

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Figure P10 World overview exported from the completed ArcGIS Pro layout.

Distance to polygon edge; zero if inside. Site presence and proximity do not establish contamination. Member precision is unverified.

All analyses use copies of 2,874 dioceses and 2,739 LSAP member points. Copies of 122 episcopal conference features and 194 country features preserve ecclesiastical context. Stable DIO_ID links outputs to the copied boundaries. Original foundational data were not edited (LSAP Pollution Atlas, 2026, foundation_manifest.json).

Analysis workflow

The following numbered sequence describes the executed ArcPy and Python analyses as a typed ModelBuilder style workflow. Custom raster, station aggregation and geometry operations are script steps, not a claim that an ArcGIS ModelBuilder model file was run. Source data: (OpenStreetMap contributors, 2026; OpenStreetMap Wiki contributors, n.d.).

1. Copy foundational members into Member_Landfill_Proximity.
2. Generate geodesic distance to nearest landfill polygon edge; members inside a footprint receive zero distance.
3. Generate all near matches within 20 km; summarize distinct sites within 1, 5 and 20 km and join to every member.
4. Classify nearest distances at 1, 5, 20 and 50 km; nearby members are darker. Retain local footprint detail and diocesan/conference context.
5. Create popups with nearest site ID, distance and buffer counts; preserve source inventory and all member attributes.

Output fields and files

Field or band	Meaning
NearSite_ID and NearSite_km	Nearest mapped site and distance to footprint edge
Sites1km Sites5km Sites20km	Distinct site counts at three distance thresholds
ProximityNote	Interpretation and positional precision note

Analysis dataset: PollutionAnalysis.gdb/Member_Landfill_Proximity. Styles/P10.lyrx and Styles/P10.mapx preserve layer and map definitions. Map_Exports/P10_World_overview.png preserves the layout preview. The editable project is LSAP_Pollution.aprx.

Interpretation and data handling

Site presence and proximity do not establish pollution or exposure. LSAP coordinates have unverified precision. Waste Atlas remains a documented supplement gap.

The working inventory contains 51,748 unique sites after removing 2,529 duplicate records. Of these, 50,996 receive a primary diocese assignment and 752 remain unassigned. Raw regional JSON, OSM_Landfill_Source and Landfills_Working remain available. Site control status and pollution severity are unknown. The European mirror is older than most other regions; retrieval date does not imply a uniform data vintage (LSAP Pollution Atlas, 2026, waste_QA.json).

Cartographic classes are descriptive display intervals, not regulatory or ecological quality grades. The pollution ramp is #E1E2DB, #B7B8A9, #898A7C, #74746A and #5D5E54. Lower oxygen and shorter proximity distances use the dark end. Foundation member dots use teal #247C80; proximity maps color member points by distance.

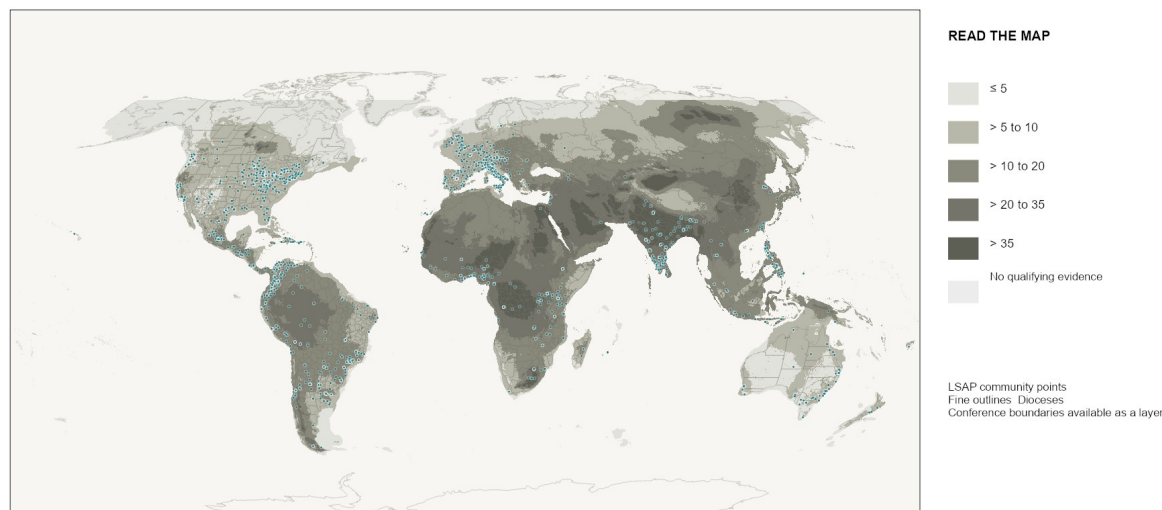
P11 Annual PM25 pollution surface

This report explains the source data, executed workflow and interpretation of annual pm25 pollution surface. It documents the existing ArcGIS Pro output, including its coverage limits and retained ecclesiastical context. (ACAG, n.d.; Shen et al., 2024).

Map P11 | Documentation date 6 October 2026 | Units $\mu\text{g}/\text{m}^3$ | Air year 2021

Annual PM2.5 pollution surface

Annual 2021 - $\mu\text{g}/\text{m}^3$



SatPM V6GL03

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Figure P11 World overview exported from the completed ArcGIS Pro layout.

Native annual concentration surface; individual cells are displayed without the diocesan 50 percent coverage display rule.

All analyses use copies of 2,874 dioceses and 2,739 LSAP member points. Copies of 122 episcopal conference features and 194 country features preserve ecclesiastical context. Stable DIO_ID links outputs to the copied boundaries. Original foundational data were not edited (LSAP Pollution Atlas, 2026, foundation_manifest.json).

Analysis workflow

The following numbered sequence describes the executed ArcPy and Python analyses as a typed ModelBuilder style workflow. Custom raster, station aggregation and geometry operations are script steps, not a claim that an ArcGIS ModelBuilder model file was run. Source data: (ACAG, n.d.; Shen et al., 2024).

1. Download SatPM V6GL03 CNN global annual 2021 NetCDF; preserve source and attribution.
2. Read native latitude/longitude coordinates and PM25 variable; export a float GeoTIFF on its native grid, units $\mu\text{g}/\text{m}^3$. Convert negative scaled source fill values to NoData in PM25_validated.tif before aggregation and display.
3. Apply the shared weighted raster aggregation to each diocesan copy; retain AirMean, AirMedian, ValidAreaPct, ValidCells and AirStory. Correct latitude-extent coverage against the full polygon.
4. Set AirDisplay to AirMean when valid area coverage is at least 50%; otherwise leave null. Join all ecclesiastical context through the copied polygon attributes.

5. Display PM25_validated.tif with breaks at 5, 10, 20 and 35 $\mu\text{g}/\text{m}^3$ above the country backdrop. Overlay copied LSAP points, diocesan outlines and optional conferences. Companion diocesan statistics are in P01; the 50 percent diocesan threshold does not mask this native surface.

Area weighting and coverage

Both concentration rasters use geographic cells, not equal area cells. For diocesan aggregation: weighted mean = $\Sigma(c_i \times a_i) / \Sigma(a_i)$, summing only valid cells. a_i is geodesic row cell area multiplied by native grid overlap fraction. This treats concentration as constant within a cell; geographic boundary fractions approximate true geodesic overlap within boundary cells. Spatial median uses the same area weights.

Coverage uses the full polygon denominator, including the PM25 extent correction at 60°S–70°N. AirDisplay requires at least 50 percent valid coverage while all computed statistics remain in attributes. The NO2 implementation uses global native grid polygon overlap for its denominator; PM25 partial extent correction uses the ratio of clipped to full geodesic polygon area. These are documented numerical approximations, not population weighting.

Output fields and files

Field or band	Meaning
Band 1	Annual concentration on the native 0.01 degree grid
NoData	Negative scaled source fill values excluded

Analysis dataset: Rasters/PM25_validated.tif. Styles/P11.lyrx and Styles/P11.mapx preserve layer and map definitions. Map_Exports/P11_World_overview.png preserves the layout preview. The editable project is LSAP_Pollution.aprx.

Interpretation and data handling

2021 is the selected common air comparison year. Class boundaries support cartography; they do not classify annual health compliance. Source: SatPM global annual file.

Credit the Atmospheric Composition Analysis Group for the SatPM V6GL03 data, and retain the source and version citation when sharing the map. The source acquisition record identifies CC BY 4.0. Source files and the validated raster copy remain in the project.

Cartographic classes are descriptive display intervals, not regulatory or ecological quality grades. The pollution ramp is #E1E2DB, #B7B8A9, #898A7C, #74746A and #5D5E54. Lower oxygen and shorter proximity distances use the dark end. Foundation member dots use teal #247C80; proximity maps color member points by distance.

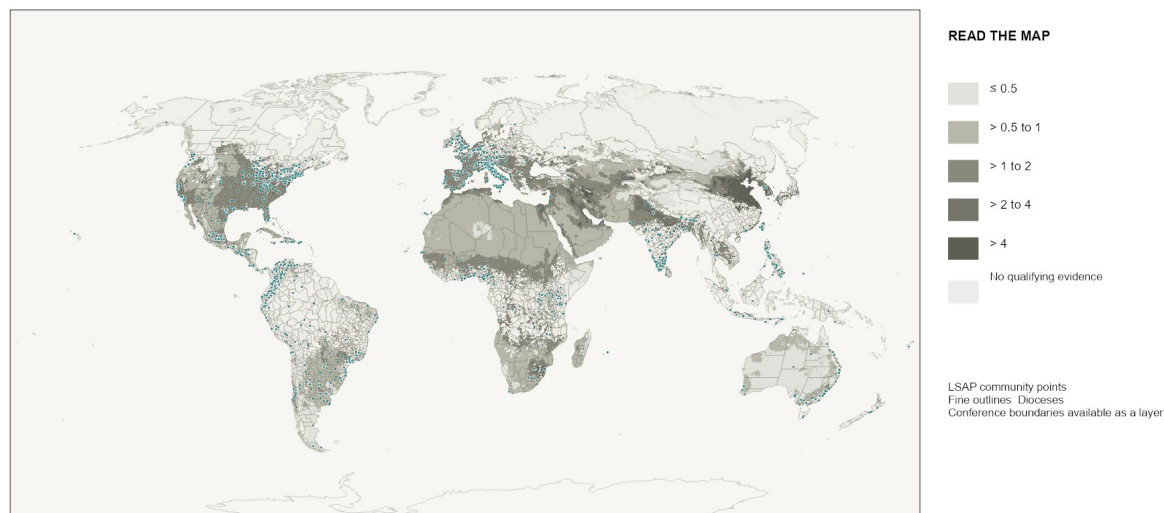
P12 Annual NO2 atmospheric composite

This report explains the source data, executed workflow and interpretation of annual no2 atmospheric composite. It documents the existing ArcGIS Pro output, including its coverage limits and retained ecclesiastical context. (Glissenaar et al., 2024, 2025; KNMI, n.d.).

Map P12 | Documentation date 6 October 2026 | Units 10^{15} molecules/cm² | Air year 2021

Annual NO2 atmospheric composite

Annual 2021 · 10^{15} molecules/cm²



KNMI ESA CCI+ TROPOMI L3 v1.26

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Figure P12 World overview exported from the completed ArcGIS Pro layout.

Native composite surface; only cells passing temporal and retrieval filters are shown.

All analyses use copies of 2,874 dioceses and 2,739 LSAP member points. Copies of 122 episcopal conference features and 194 country features preserve ecclesiastical context. Stable DIO_ID links outputs to the copied boundaries. Original foundational data were not edited (LSAP Pollution Atlas, 2026, foundation_manifest.json).

Analysis workflow

The following numbered sequence describes the executed ArcPy and Python analyses as a typed ModelBuilder style workflow. Custom raster, station aggregation and geometry operations are script steps, not a claim that an ArcGIS ModelBuilder model file was run. Source data: (Glissenaar et al., 2024, 2025; KNMI, n.d.).

1. Download all twelve 2021 KNMI ESA CCI+ TROPOMI L3 v1.26 monthly files (fv1.0 filename version), native 900 × 1800 grid.
2. Select monthly cells with qa_L3=1, land/transition mask, finite column value and fractional effective monthly coverage × calendar days ≥ 10. This effective coverage is not a literal count of valid days.
3. Calculate calendar-day-weighted means of qualifying monthly columns. Require at least nine qualifying months and representation from all four quarters. Preserve negative retrieval values; divide by 10^{15} to express molecules/cm² in readable units.
4. Export annual concentration and qualifying-month-count rasters; audit both CRS and grids. Aggregate annual concentration using geodesic cell areas and fractional overlaps. Retain weighted spatial median and full-polygon valid coverage.

5. Display NO2_Annual_2021.tif with breaks at 0.5, 1, 2 and 4×10^{15} molecules/cm² above the country backdrop. Overlay copied LSAP points, diocesan outlines and optional conferences. Companion diocesan statistics are in P02.

Area weighting and coverage

Both concentration rasters use geographic cells, not equal area cells. For diocesan aggregation: weighted mean = $\sum(c_i \times a_i) / \sum(a_i)$, summing only valid cells. a_i is geodesic row cell area multiplied by native grid overlap fraction. This treats concentration as constant within a cell; geographic boundary fractions approximate true geodesic overlap within boundary cells. Spatial median uses the same area weights.

Coverage uses the full polygon denominator, including the PM25 extent correction at 60°S–70°N. AirDisplay requires at least 50 percent valid coverage while all computed statistics remain in attributes. The NO2 implementation uses global native grid polygon overlap for its denominator; PM25 partial extent correction uses the ratio of clipped to full geodesic polygon area. These are documented numerical approximations, not population weighting.

Output fields and files

Field or band	Meaning
Band 1	Annual qualifying monthly composite at native 0.2 degree grid
NO2_ValidMonths_2021.tif	Separate qualifying month count raster retained

Analysis dataset: Rasters/NO2_Annual_2021.tif. Styles/P12.lyrx and Styles/P12.mapx preserve layer and map definitions. Map_Exports/P12_World_overview.png preserves the layout preview. The editable project is LSAP_Pollution.aprx.

Interpretation and data handling

A cell with nine qualifying months is an observed-month estimate, not complete annual sampling. Tropospheric column density is different from surface concentration or personal exposure. KNMI source.

Cartographic classes are descriptive display intervals, not regulatory or ecological quality grades. The pollution ramp is #E1E2DB, #B7B8A9, #898A7C, #74746A and #5D5E54. Lower oxygen and shorter proximity distances use the dark end. Foundation member dots use teal #247C80; proximity maps color member points by distance.

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